Deterministic Optimization

Linear Programming Duality

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Complementarity Slackness

LP Duality

Learning Objectives for this lesson

- Examine complementarity slackness, an equivalent property to strong duality

Complementarity Slackness

Theorem 1 (Complementary Slackness)**.** *Let* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *be feasible solutions to the primal and dual problem, respectively. Then* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *are optimal solutions for the two respective problems if and only if they satisfy the following conditions:*

> *Primal Complementary Slackness:* $y_i(\boldsymbol{a}_i^\top \boldsymbol{x} - b_i) = 0$ *for all* i*. In words, either the* i*-th primal constraint is active (binding, tight) so* $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i$*, or the corresponding dual variable* $y_i = 0$*.*
>
> *Dual Complementary Slackness:* $x_j(c_j - \boldsymbol{y}^\top \boldsymbol{A}_j) = 0$ *for all* j*. In words, either the* j*-th dual constraint is active (binding, tight) so* $\boldsymbol{y}^\top \boldsymbol{A}_j = c_j$*, or the corresponding primal variable* $x_j = 0$*.*

Example 1

- Consider the following LP primal and dual pair

$$
\begin{aligned}
\min \quad & 13x_1 + 10x_2 + 6x_3 \\
\text{s.t.} \quad & 5x_1 + x_2 + 3x_3 = 8 \\
& 3x_1 + x_2 = 3 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
\qquad\qquad
\begin{aligned}
\max \quad & 8y_1 + 3y_2 \\
\text{s.t.} \quad & 5y_1 + 3y_2 \leq 13 \\
& y_1 + y_2 \leq 10 \\
& 3y_1 \leq 6.
\end{aligned}
$$

- Dual problem only has 2 variables and can be solved graphically
    - Dual optimal solution: $y_1^* = 2, y_2^* = 1$
- Can we get primal solution using this information but without solving the primal problem?

Example 1 (cont.)

1. Use Dual Complementary Slackness: At the dual optimum, we have

$$\begin{aligned} 5y_1^* + 3y_2^* &= 13 \\ y_1^* + y_2^* &= 3 < 10 \\ 3y_1^* &= 6. \end{aligned}$$

   Therefore, by the Dual Complementary Slackness $x_2^*(y_1^* + y_2^* - 10) = 0$, we know $x_2^* = 0$.

2. Primal Complementary Slackness is automatically satisfied, because the primal constraints are equalities. From above, we know $x_2^* = 0$, therefore we have

$$\begin{aligned} 5x_1^* + 3x_3^* &= 8 \\ 3x_1^* &= 3. \end{aligned}$$

   The optimal primal solution is $x_1^* = 1, x_2^* = 0, x_3^* = 1$.

Example 2

- Consider the same primal and dual problems in Example 1.
- Suppose we know primal optimal solution $x_1^* = 1, x_2^* = 0, x_3^* = 1$

1. Use Dual Complementary Slackness: Since both x_1^* and x_3^* are nonzero, in order to have $x_1^*(5y_1^* + 3y_2^* - 13) = 0$, and $x_3^*(3y_1^* - 6) = 0$, we must have

$$5y_1^* + 3y_2^* = 13$$
$$3y_1^* = 6,$$

   which gives $y_1^* = 2, y_2^* = 1$. This is a feasible dual solution.

2. Check Primal Complementary Slackness: Since the primal constraints are equalities, Primal Complementary Slackness automatically holds for this pair of primal and dual solutions. Thus, by the Complementary Slackness Theorem, we know $(x_1^* = 1, x_2^* = 0, x_3^* = 1)$ and $(y_1^* = 2, y_2^* = 1)$ are primal and dual optimal.

Summary

- We learned:
  - Complementarity slackness
  - How to use it to quickly solve LP